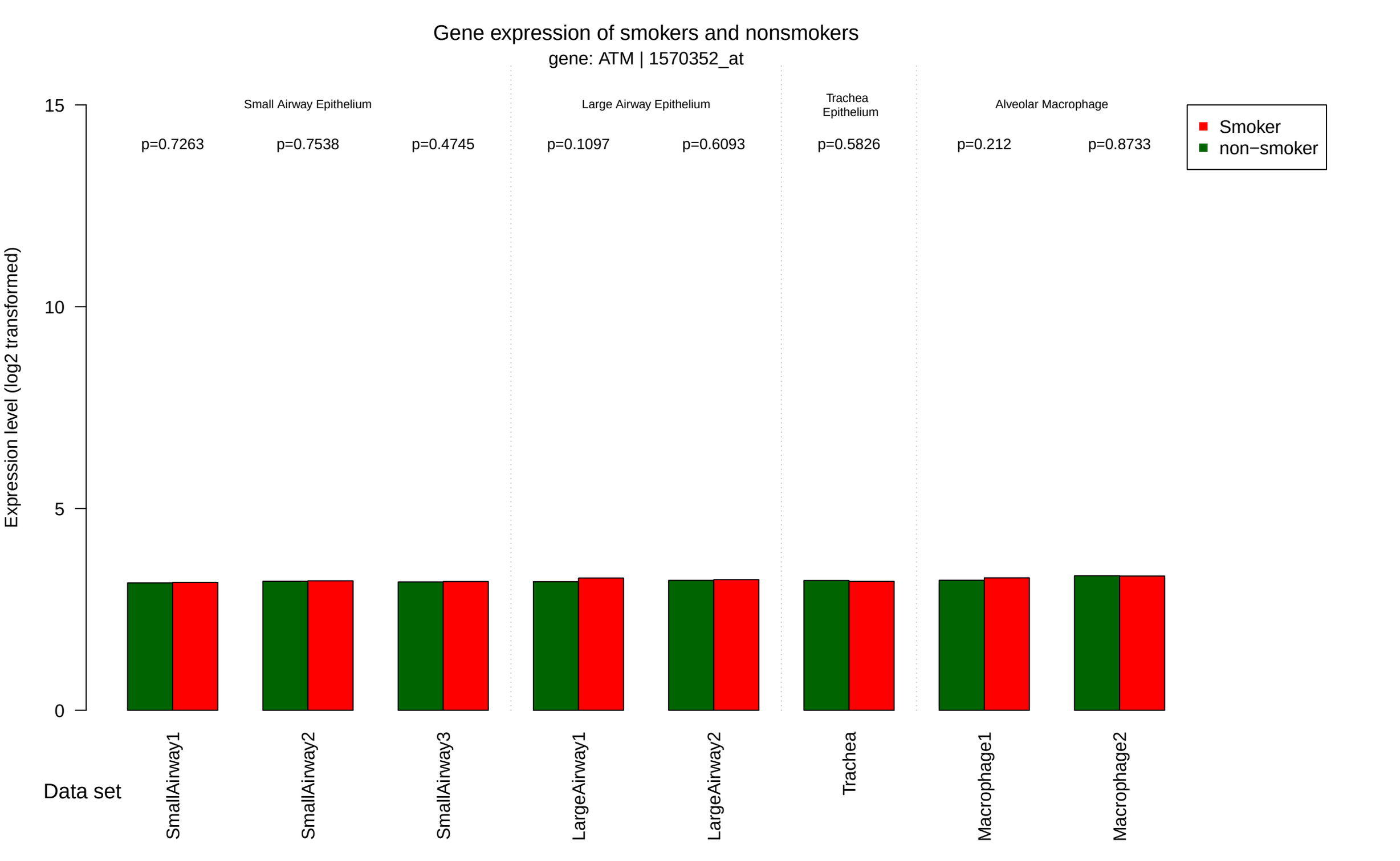
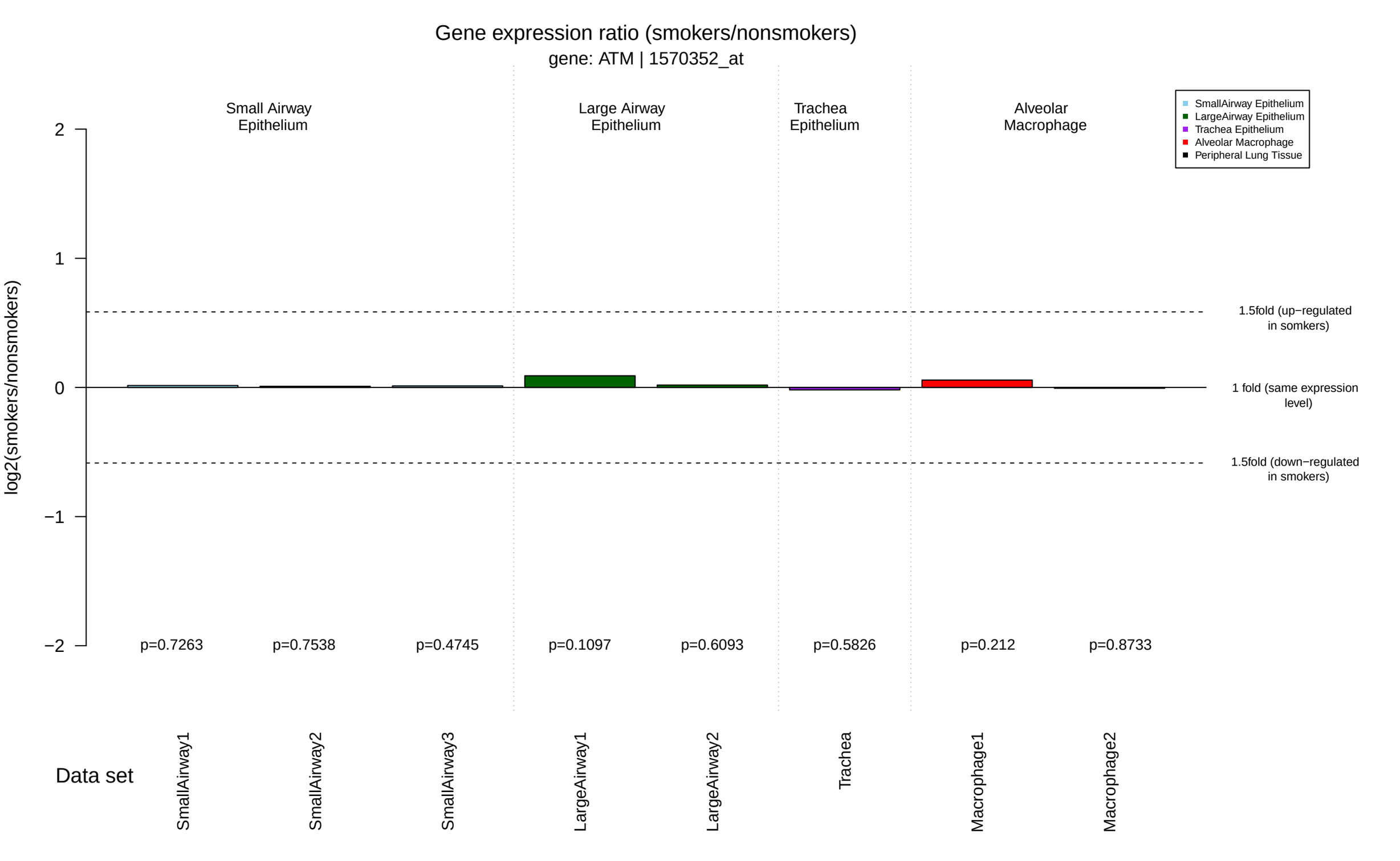
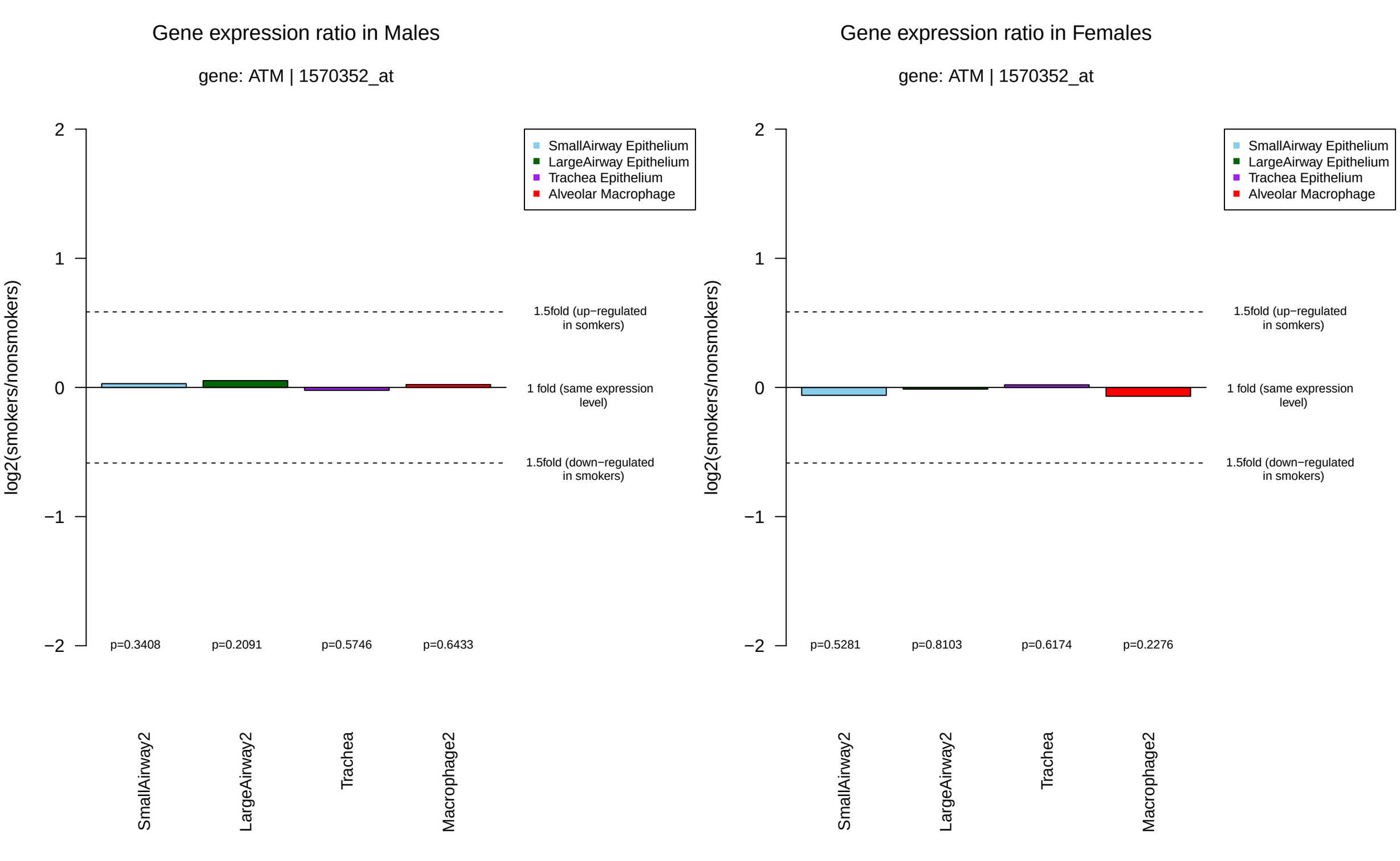
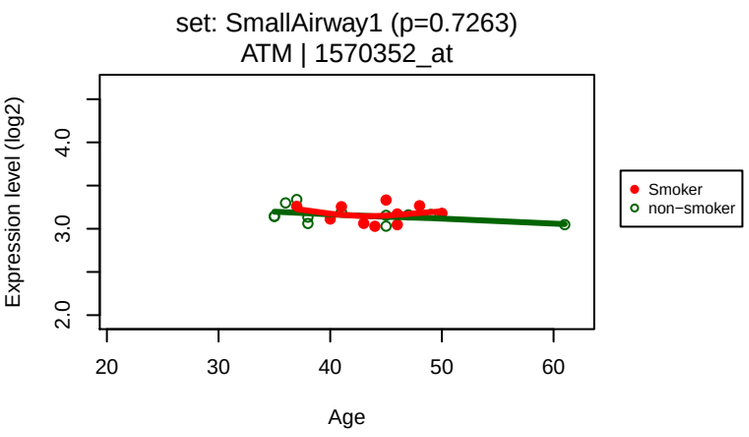
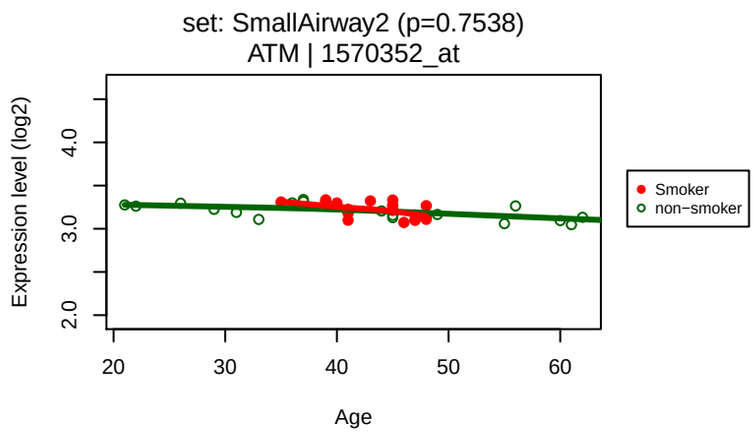
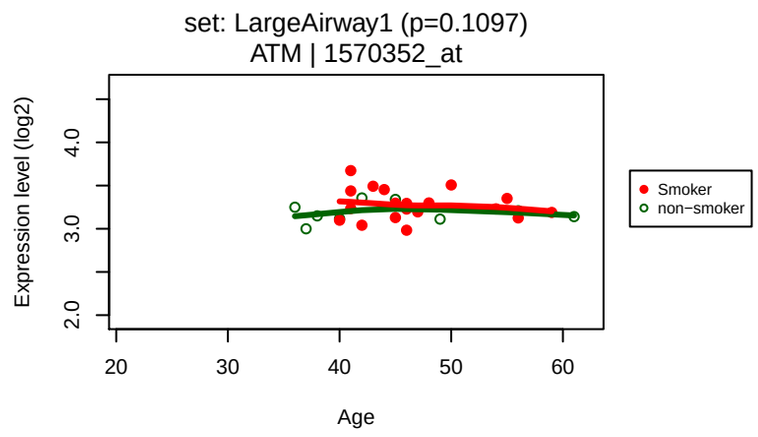

Gene expression of smokers and nonsmokers
gene: ATM | 1570352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7263
p=0.7538
p=0.4745
p=0.1097
p=0.6093
p=0.5826
p=0.212
p=0.8733
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

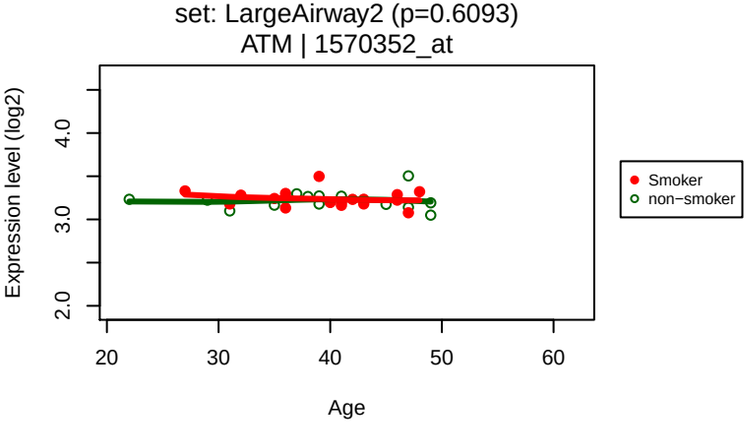

Gene expression ratio (smokers/nonsmokers)
gene: ATM | 1570352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7263
p=0.7538
p=0.4745
p=0.1097
p=0.6093
p=0.5826
p=0.212
p=0.8733
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

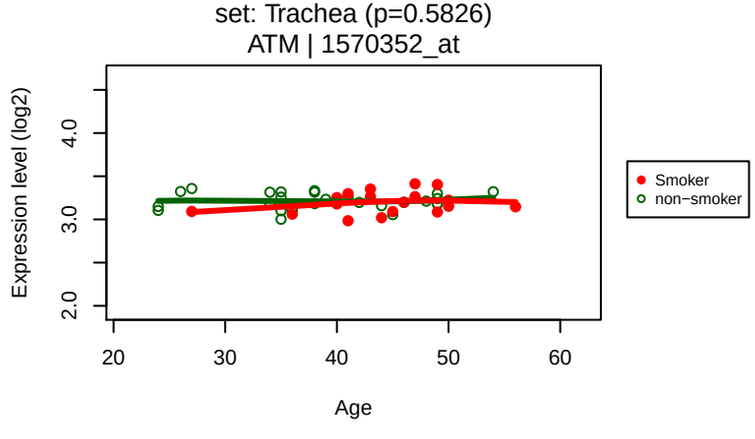

Gene expression ratio in Males
gene: ATM | 1570352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3408
p=0.2091
p=0.5746
p=0.6433
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATM | 1570352_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5281
p=0.8103
p=0.6174
p=0.2276
SmallAirway2
LargeAirway2
Trachea
Macrophage2

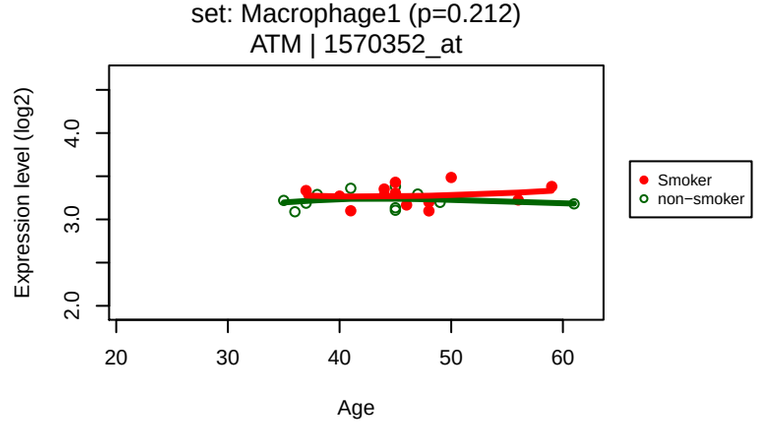

set: SmallAirway1 (p=0.7263)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker

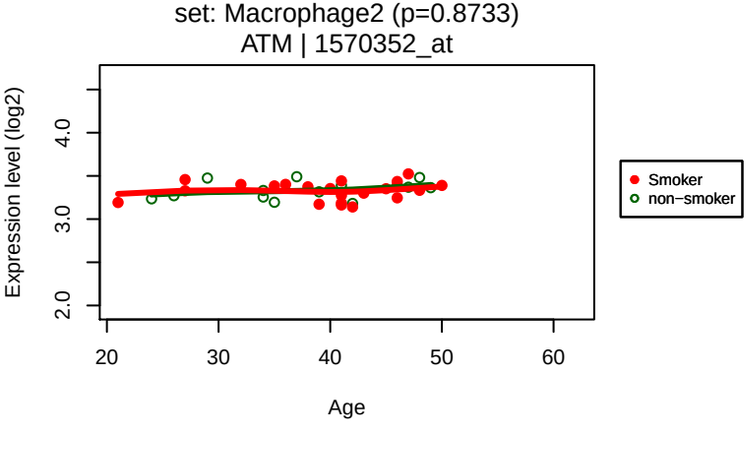

set: SmallAirway2 (p=0.7538)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker

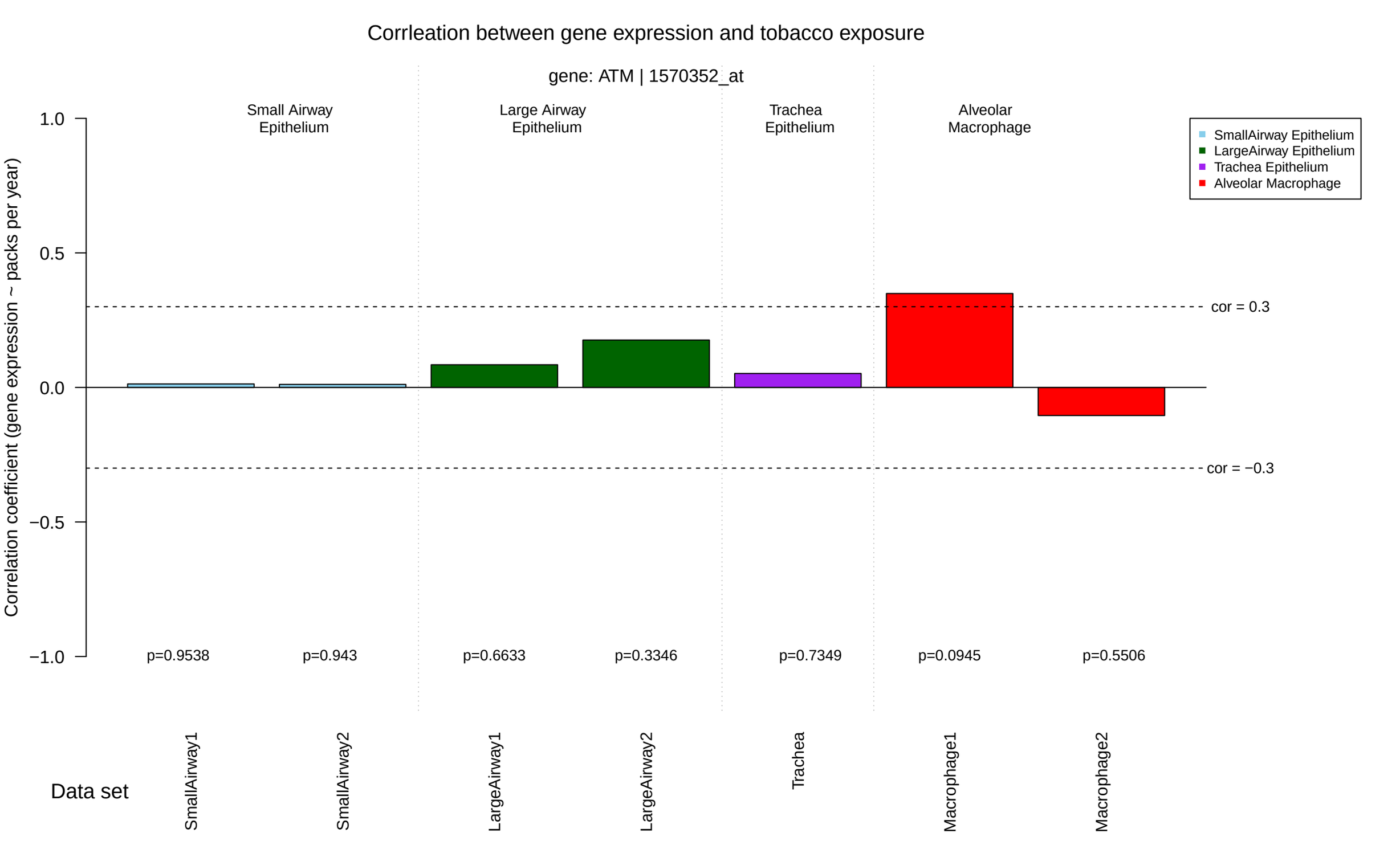

set: LargeAirway1 (p=0.1097)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6093)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5826)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.212)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8733)
ATM | 1570352_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATM | 1570352_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9538
p=0.943
p=0.6633
p=0.3346
p=0.7349
p=0.0945
p=0.5506
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2